T4 Checkpoint 1: Downloaded data

Group T4 (CPSC 4430), data start owner Jackson Mahr. Code repository: https://github.com/JacksonMahr782/Machine-Learning. Raw files are kept in Box (data/raw/), not GitHub.

Table 1. Every downloaded file

#	File	Source	Release / version	Format	Size (as shown in Windows)	Date downloaded	One row is	One column is	Used for
1	CRISPRGeneEffect.csv	DepMap Portal	Public 26Q3	CSV	485,489 KB (~474 MB)	2026-10-15	One cell line (ModelID)	One gene ('SYMBOL (ENTREZ)')	Label only (Chronos effect)
2	Model.csv	DepMap Portal	Public 26Q3	CSV	772 KB	2026-10-15	One cell line	One attribute (ModelID, OncotreeCode, lineage...)	Picking LUAD lines (OncotreeCode = LUAD)
3	TCGA-LUAD.star_tpm.tsv.gz	UCSC Xena (GDC TCGA-LUAD hub)	star_tpm version not recorded	TSV (gzip)	164,516 KB (~161 MB)	2026-10-19	One gene (Ensembl ID)	One sample (tumour or normal)	Features: expression, enrichment, variability
4	TCGA-LUAD.clinical.tsv.gz	UCSC Xena (GDC TCGA-LUAD hub)	phenotype v09-07-2024	TSV (gzip)	147 KB	2026-10-19	One sample	One clinical/phenotype field	Tumour vs normal sample type
5	luad_tcga_pan_can_atlas_2018.tar.gz (data_cna.txt inside)	cBioPortal	TCGA LUAD PanCancer Atlas 2018	TAR.GZ (tab-separated text inside)	912,219 KB (~891 MB)	2026-10-19	One gene (Hugo_Symbol)	One tumour sample (GISTIC -2, -1, 0, +1, +2)	Features: amplification frequency
6	subcellular_location.tsv.zip	Human Protein Atlas	Ensembl 109	TSV (zip)	247 KB	2026-10-19	One gene	Gene, Main/Additional location, Reliability	Surface eligibility + split stratification
7	E-MTAB-16585.zip	ArrayExpress (TROP2 paper)	E-MTAB-16585	ZIP of processed .txt files	1,304,207 KB	2026-10-19	(12 processed .txt files across the 3 zips)	(cells x genes + cell metadata)	Paper reproduction (Figure 1)
8	E-MTAB-16585-2.zip	ArrayExpress (TROP2 paper)	E-MTAB-16585	ZIP of processed .txt files	1,281,957 KB	2026-10-19	(see row 7)	(see row 7)	Paper reproduction (Figure 1)
9	E-MTAB-16585 3.zip	ArrayExpress (TROP2 paper)	E-MTAB-16585	ZIP of processed .txt files	1,076,004 KB	2026-10-19	(see row 7)	(see row 7)	Paper reproduction (Figure 1)
10	figures/Figure_01.ipynb	GitHub ORCA-HD/TROP2_in_CRC	commit d04c1bd (2026-07-01)	Jupyter notebook	592 KB	2026-10-19 (viewed)	-	-	Paper reproduction code

All 12 processed E-MTAB-16585 .txt files were confirmed present across the three zips on 2026-10-21. Screenshots proving each download are in proof/.

Table 2. Layout of the gene table we will build at Checkpoint 2 (example rows, not real values)

gene_symbol	expr_mean_tumor	expr_enrichment	expr_sd_tumor	gistic_amp_freq	gistic_highamp_freq	is_surface	n_luad_lines	label_dependency	split
GENE_A	6.2	1.4	1.1	0.18	0.03	TRUE	11	0.42	train
GENE_B	3.0	-0.5	0.8	0.05	0.00	FALSE	11	0.05	validation
GENE_C	8.7	2.3	1.6	0.31	0.07	TRUE	9	0.88	test
GENE_D	1.1	0.2	0.4	0.02	0.00	FALSE	2		(unlabeled)

One row per gene. Features come from Xena and cBioPortal; the label comes only from DepMap; is_surface comes from HPA and is used only for eligibility and the stratified split. GENE_D has fewer than 3 LUAD lines, so it is unlabeled and left out of all splits.